Code No: 117JV **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
INFORMATION SECURITY ASSESSMENTS AND AUDITS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	State objectives of information security audit.	[2]
b)	Write about Storage devices.	[3]
c)	What is mean by result analysis?	[2]
d)	Write characteristics of firewall security audit.	[3]
e)	What are challenges in remediation?	[2]
f)	Write short notes on threats.	[3]
g)	Define vulnerability assessment.	[2]
h)	What are the steps for risk assessment?	[3]
i)	List limitations of configuration management.	[2]
j)	What are the typical components of testing configuration?	[3]

PART – B

(50 Marks)

2.a)	Discuss in detail about Phases of Information Security Audit and Strategies.	
b)	Illustrate Ethics of an Information Security Auditor.	[5+5]
	OR	
3.a)	Compare black box and white box information security methodologies.	
b)	Describe in detail about the Infrastructure, Networks and Communication routes.	[5+5]
4.a)	Elaborate implementations issues regarding internal and external security Audit.	
b)	State the benefits and objectives of information security audit.	[5+5]
	OR	
5.a)	Explain the system-level, application level and user information security audit.	
b)	Define Pre-Audit checklist? Explain post auditing actions.	[5+5]
6.a)	Explain the concept of Computer-based Social Engineering in vulnerability Management.	
b)	Discuss vulnerabilities in information security.	[5+5]
	OR	
7.a)	Discuss in detail about vulnerability scanning.	
b)	What are the steps included in developing vulnerability testing policies?	[5+5]

8.a) Compare the quantitative and qualitative information security risk assessment approaches.
b) With neat diagram, explain the risk management hierarchy. [5+5]

OR

9.a) Write the notes on Vulnerability Assessment Reports – Tools.
b) Explain Characteristics of a Good Vulnerability Assessment Solutions and Considerations. [5+5]

10. Explain about Configuration Management Requirements. [10]

OR

11. Write about Development of configuration Control Policies. [10]

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B. Tech IV Year I Semester Examinations, 2023
INFORMATION SECURITY ASSESSMENTS AND AUDITS
( )

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) What are the security objectives information security methodologies. [2]
b) Summarize the information security strategies. [3]
c) What is mean by Pre-Audit checklist? [2]
d) Write objectives of Web application security auditing. [3]
e) Define Vulnerability testing? [2]
f) Write short notes on threat management. [3]
g) What is Residual Risk? [2]
h) What are the major limitations of Good Vulnerability Assessment? [3]
i) Summarize the major considerations for Configuration Control Policies. [2]
j) What is the need for Plan-Control? [3]

PART – B

(50 Marks)

Unit-I

2.a) Write importance of Information Security performance metrics
b) Discuss the various functions of infrastructure and network?

OR

3.a) Discuss the legal, ethical and professional issues related to Information Security.
b) Illustrate security server and storage devices in detail.

Unit-II

4.a) What are the various functions of IDS technologies in security auditing?
b) Define Result Analysis? Write about the concept of Information Standard Formats?

OR

5.a) Explain in detail about the concept of social engineering security auditing.
b) Write short note on application firewalls.

Unit-III

6.a) Describe in detail about the Information Security Vulnerabilities.

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b) Explain in detail about the stages of Vulnerability analysis.

OR

7.a) Discuss the concept of Human-based Social Engineering in vulnerability Management.

b) Elaborate the importance of remediation in vulnerability Management.

Unit-IV

8.a) List and discuss the different risk assessment approaches?

b) Discuss the important implementation issues for information security assessment systems.

OR

9.a) Define classification? Explain the types of Types of Vulnerability Assessment.

b) Write the notes on Vulnerability Risk acceptance.

Unit-V

10. Describe in detail about the Testing Configuration Management.

OR

11. Explain various criteria used by the system to determine configuration reviews?

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